F. Money Trees

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1300 | Tags: binary search, greedy, math, two pointers | https://codeforces.com/problemset/problem/1873/F

Luca is in front of a row of n trees. The i-th tree has a_i fruit and height h_i.

He wants to choose a contiguous subarray of the array $[h_l, h_{l+1}, \ldots, h_r]$ such that for each i ($l \le i$ lt r), $\boldsymbol{h_i}$ **is divisible**[dagger] **by** $\boldsymbol{h_{i+1}}$. He will collect all the fruit from each of the trees in the subarray (that is, he will collect $a_l + a_{l+1} + \ldots + a_r$ fruits). However, if he collects more than k fruits in total, he will get caught.

What is the maximum length of a subarray Luca can choose so he doesn't get caught?

[dagger] x is *divisible* by y if the ratio x/y is an integer.

Input

The first line contains a single integer t ($1 \le t \le 1000$) — the number of test cases.

The first of each test case line contains two space-separated integers n and k ($1 \le n \le 2 \cdot 10^5$; $1 \le k \le 10^9$) — the number of trees and the maximum amount of fruits Luca can collect without getting caught.

The second line of each test case contains n space-separated integers a_i ($1 \le a_i \le 10^4$) — the number of fruits in the i-th tree.

The third line of each test case contains n space-separated integers h_i ($1 \le h_i \le 10^9$) — the height of the i-th tree.

The sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case output a single integer, the length of the maximum length contiguous subarray satisfying the conditions, or 0 if there is no such subarray.

Example

Input

```
5
5 12
3 2 4 1 8
4 4 2 4 1
4 8
5 4 1 2
6 2 3 1
3 12
7 9 10
2 2 4
1 10
11
1
7 10
2 6 3 1 5 10 6
72 24 24 12 4 4 2
```

Output
3 2 1 0 3

Note

In the first test case, Luca can select the subarray with l=1 and r=3.

In the second test case, Luca can select the subarray with l=3 and r=4.

In the third test case, Luca can select the subarray with l=2 and r=2.